

Lecture 5

Baseband digital transmission

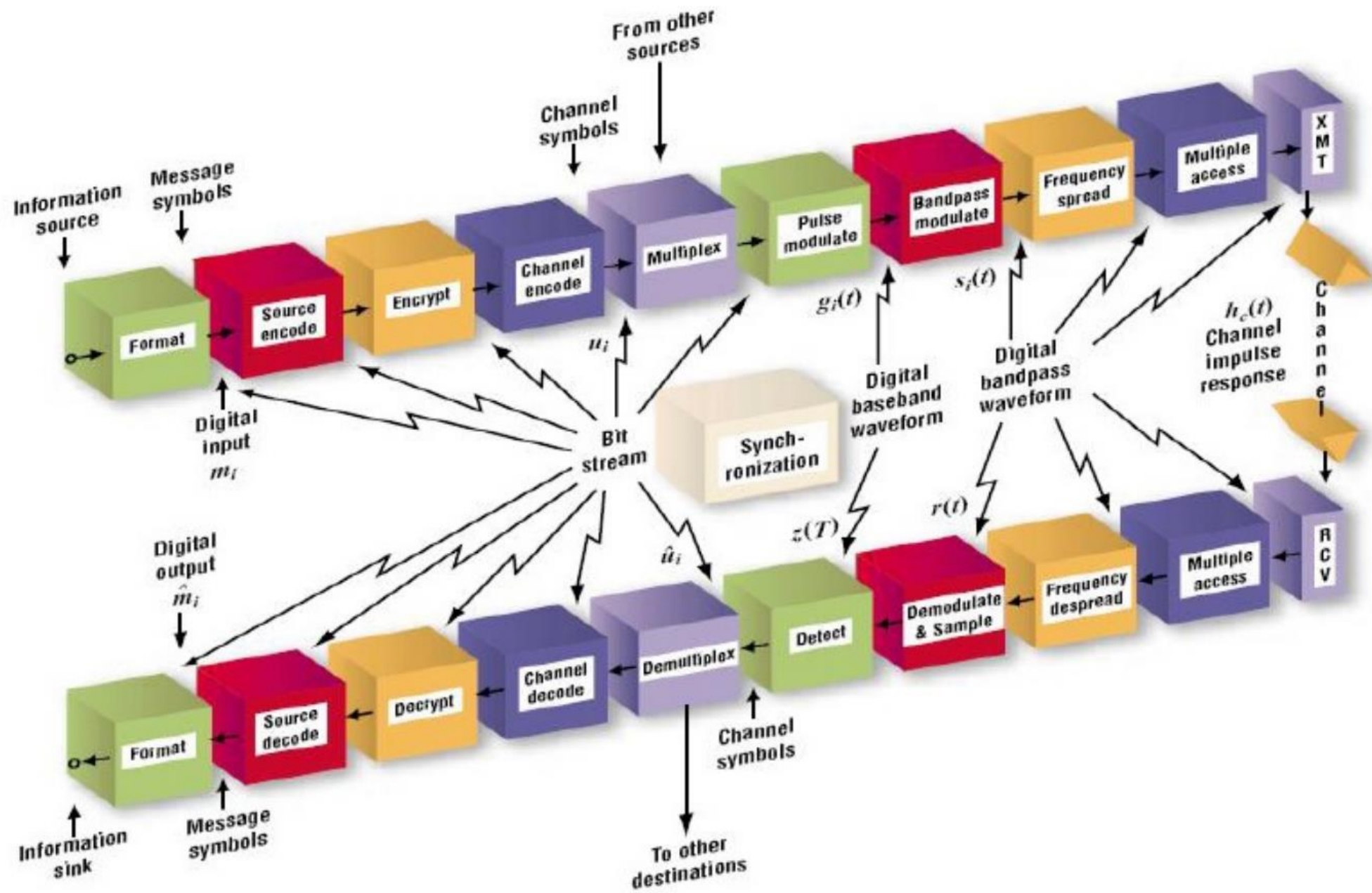
Introduction

- Generally the digital data transmission takes place in two ways.
 1. Base Band Pulse Transmission.
 2. Pass Band Pulse Transmission.

Baseband Data Transmission: The digital data is transmitted over the channel directly. There is no carrier or any modulation. This is suitable for transmission over short distances.

Passband Data Transmission: The digital data can be transmitted over the channel by using modulation techniques. This one can be used for the longer distance communication.

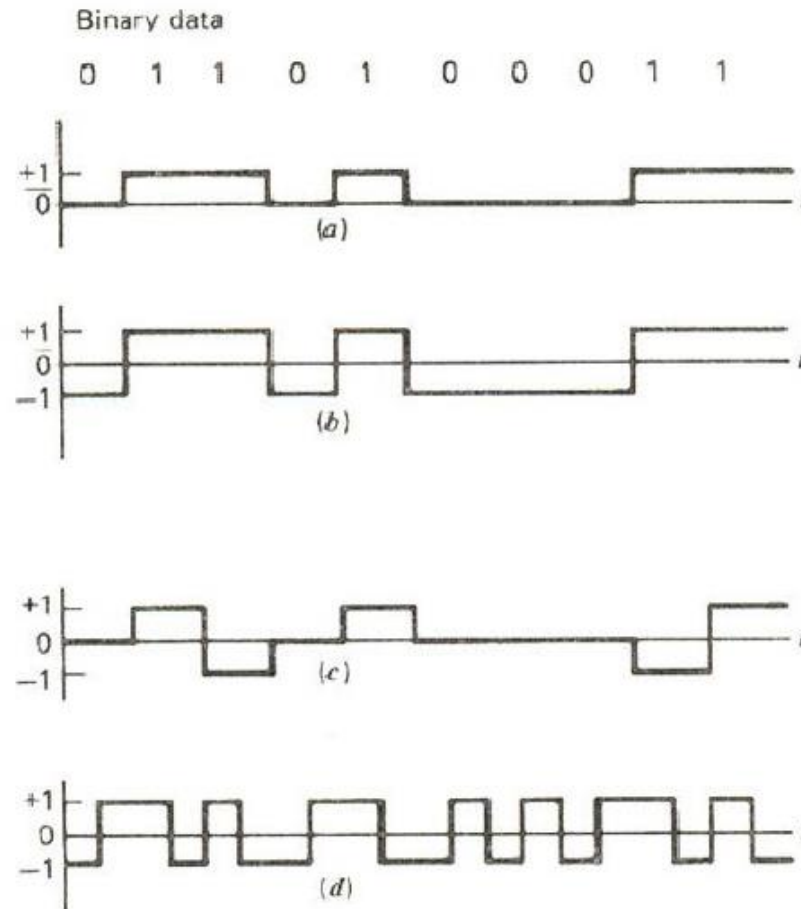
- In baseband transmission, the data bits are directly converted into signals. Generally a higher voltage level represents the bit 1, while a lower voltage level represents bit 0.
- Baseband transmission is the simplest form for the communication of information. Discrete information is communicated with specific symbols selected from a finite set of symbols.
- In baseband transmission, symbols are simply communicated as a pulse with a discrete voltage level and, for binary transmission, only two voltages are used.



Discrete PAM Signals

- ❖ The use of an appropriate for baseband representation of digital is basic to its transmission from a source to a destination
- ❖ There are some different formats for the representation of the binary data sequence
 - Unipolar format (on-off signaling)
 - Polar format
 - Bipolar format (also known as pseudoternary signaling)
 - Manchester format (also known as biphase baseband signaling)

Binary Data



Binary data formats. (a) Nonreturn-to-zero unipolar format. (b) Nonreturn-to-zero polar format. (c) Nonreturn-to-zero bipolar format. (d) Manchester format.

Data Formats

NRZ format	Coefficient A_k	Basic pulse $v(t)$
Unipolar	$A_k = \begin{cases} a, & \text{symbol 1} \\ 0, & \text{symbol 0} \end{cases}$	$v(t)$ consists of rectangular pulse of unit amplitude and duration T_b .
Polar	$A_k = \begin{cases} a, & \text{symbol 1} \\ -a, & \text{symbol 0} \end{cases}$	
Bipolar	$A_k = \begin{cases} a, -a, & \text{alternating 1s} \\ 0, & \text{symbol 0} \end{cases}$	
Manchester	$A_k = \begin{cases} a, & \text{symbol 1} \\ -a, & \text{symbol 0} \end{cases}$	$v(t)$ consists of doublet pulse of heights ± 1 , and total duration T_b .
Polar quaternary (natural coding)	$A_k = \begin{cases} 3a, & \text{dibit 11} \\ a, & \text{dibit 10} \\ -a, & \text{dibit 01} \\ -3a, & \text{dibit 00} \end{cases}$	$v(t)$ consists of rectangular pulse of unit amplitude and duration $2T_b$.

* In Figs. 6.1 and 6.2, the amplitude a is given as one.

Inter Symbol Interference

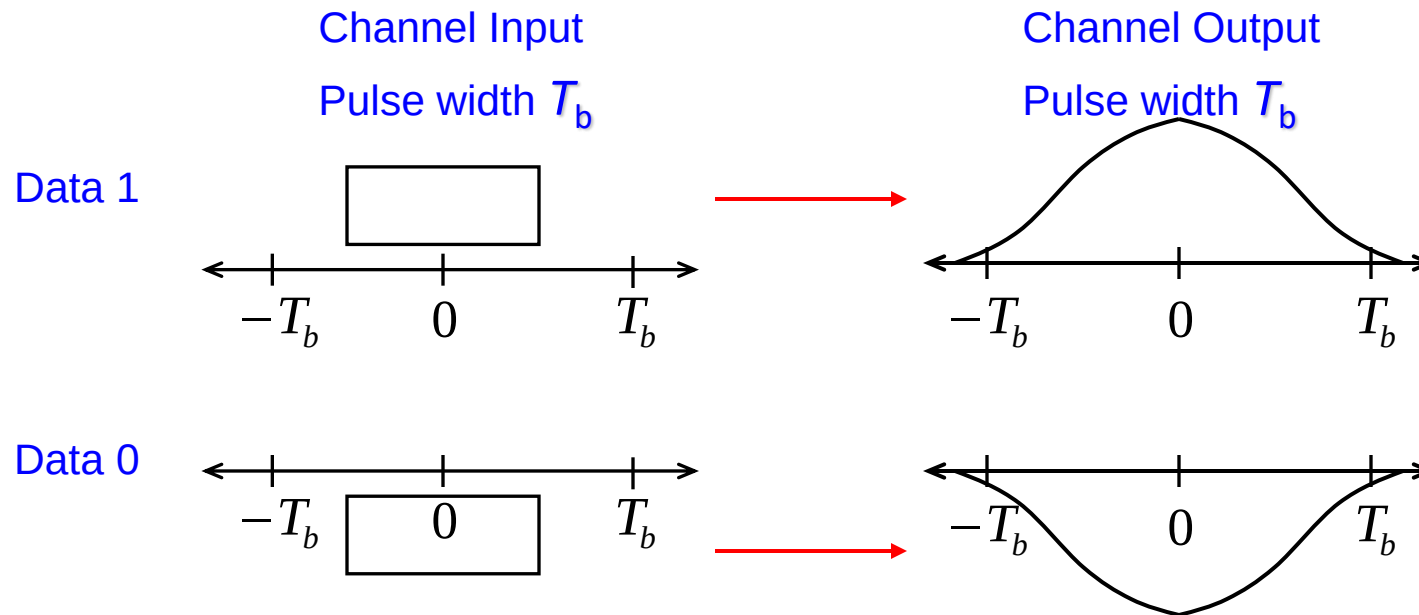
- Transmission of digital data (bit stream) over a noisy baseband channel typically suffers two channel limitations
 - ❖ Inter symbol interference (ISI)
 - ❖ Background noise (e.g., AWGN)

In case the channel is dispersive then the each received pulse is affected somewhat by adjacent pulses. This type of interference is called Inter Symbol Interference (ISI). This is the major source of bit errors. This one can be controlled by maintaining certain pulse shape in the overall system.

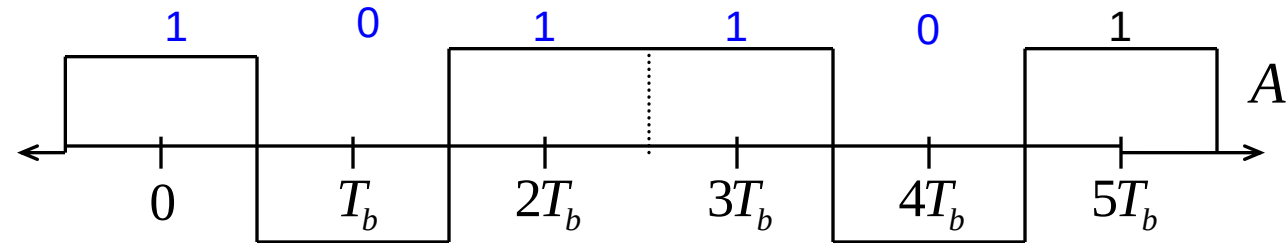
Inter Symbol Interference

❖ **Intersymbol interference (ISI)** occurs when a pulse spreads out in such a way that it interferes with adjacent pulses *at the sample instant*.

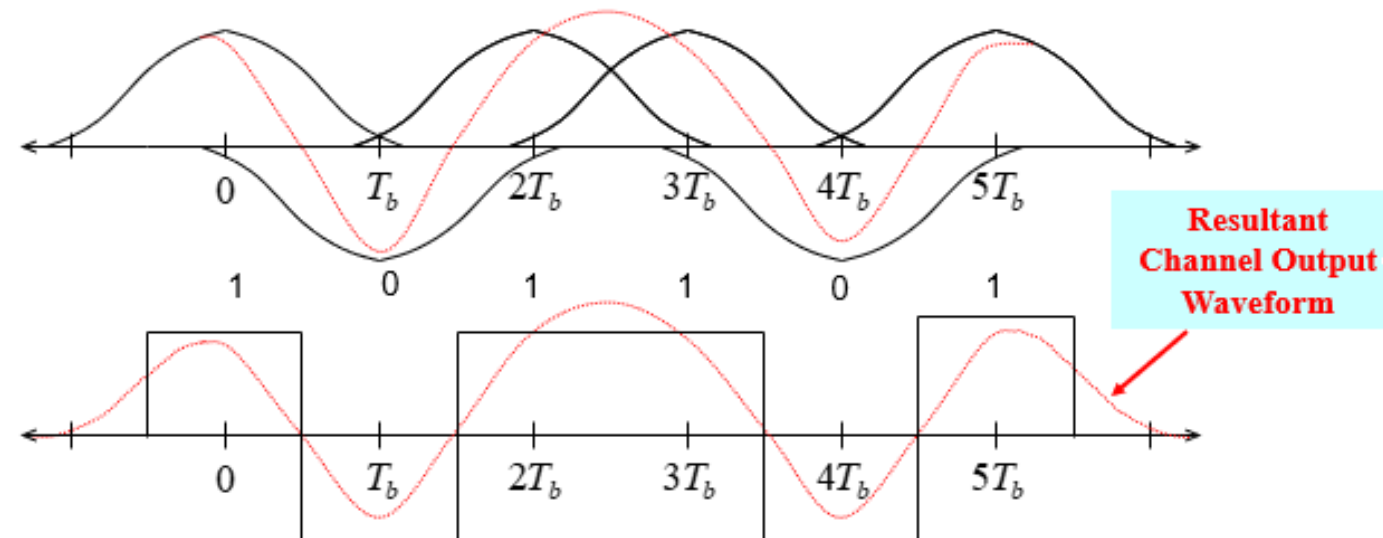
❖ Example: assume polar NRZ line code. The channel outputs are shown as spreaded (width T_b becomes $2T_b$) pulses shown (Spreading due to bandlimited channel characteristics).



❖ For the input data stream:

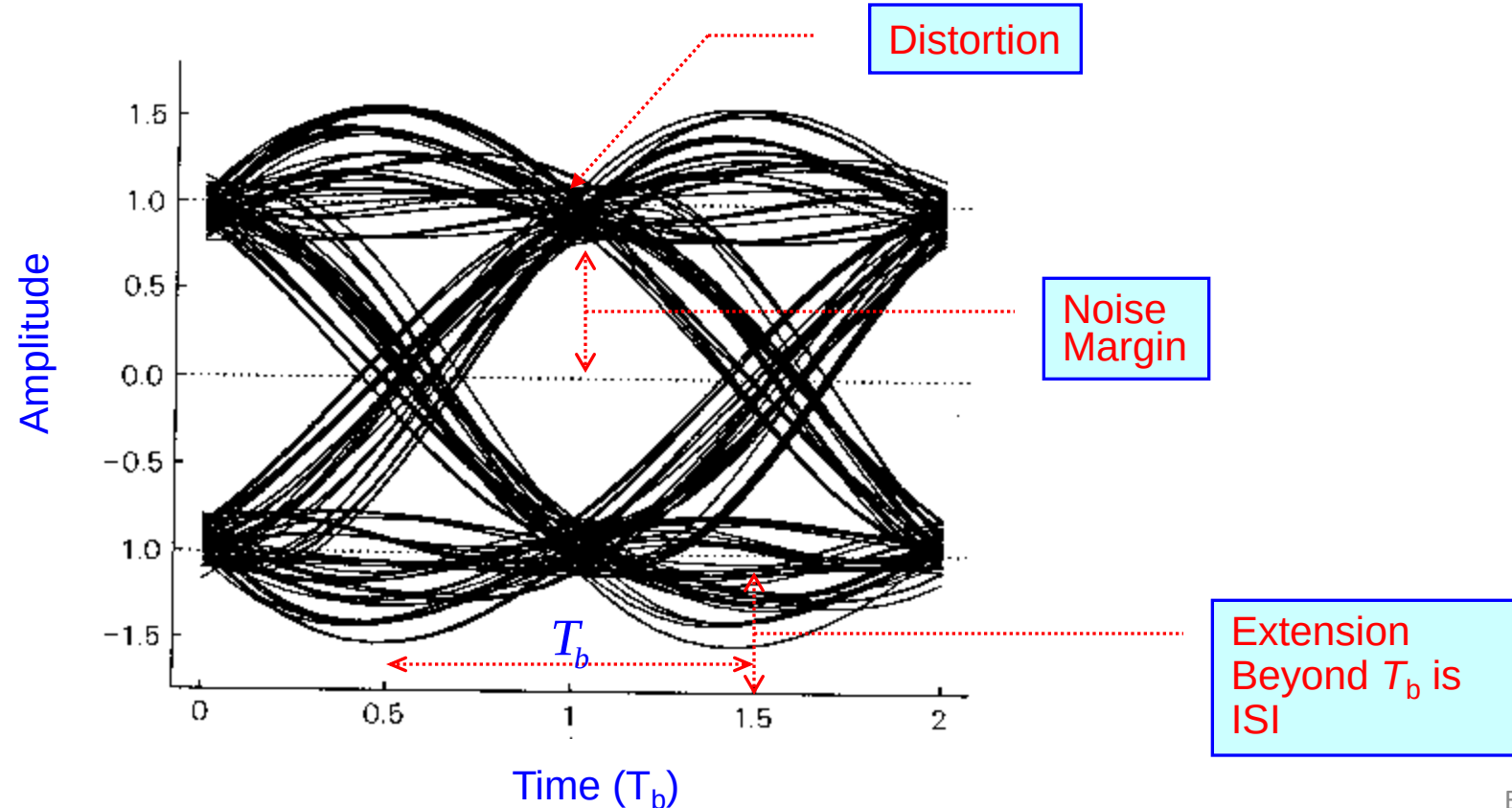


❖ The channel output is the superposition of each bit's output:

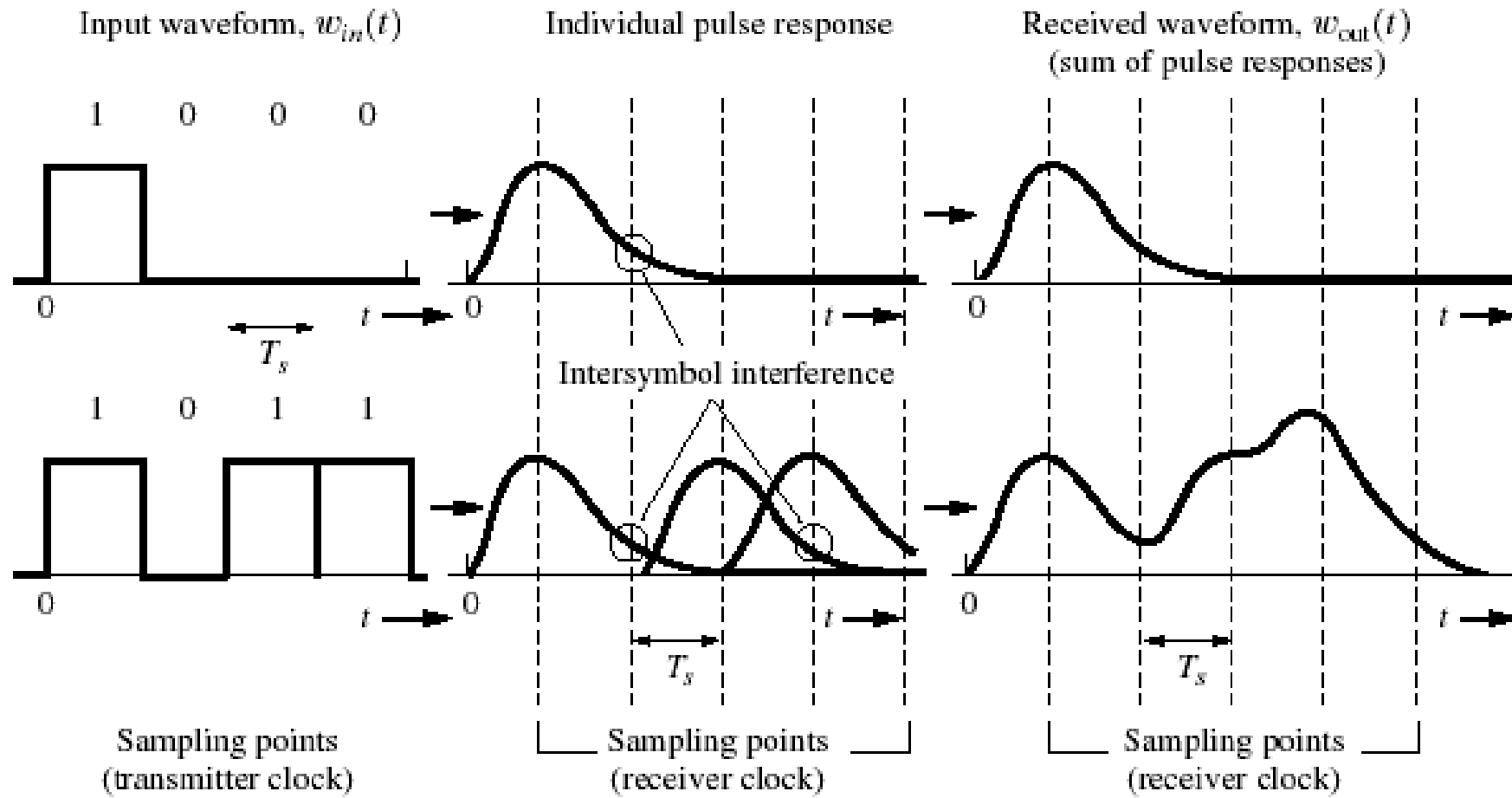


ISI on Eye Patterns

- The amount of ISI can be seen on an oscilloscope using an **Eye Diagram** or **Eye pattern**.



- If the rectangular multilevel pulses are filtered improperly as they pass through a communications system, they will spread in time, and the pulse for each symbol may be smeared into adjacent time slots and cause **Intersymbol Interference**.



Theory of ISI

❖ Consider basic elements of a baseband binary PAM system

- The input signal consists of a binary data sequence $\{b_k\}$ with a bit duration of T_b seconds
- This sequence is applied to a pulse generator, producing the discrete PAM signal

$$x(t) = \sum_{k=-\infty}^{\infty} a_k v(t - kT_b)$$

- $v(t)$ denotes the basic pulse, normalize such that $v(0) = 1$
- The coefficient a_k depends on the input data and the type of format used
- The waveform $x(t)$ represents one realization of the random process $X(t)$

Theory of ISI

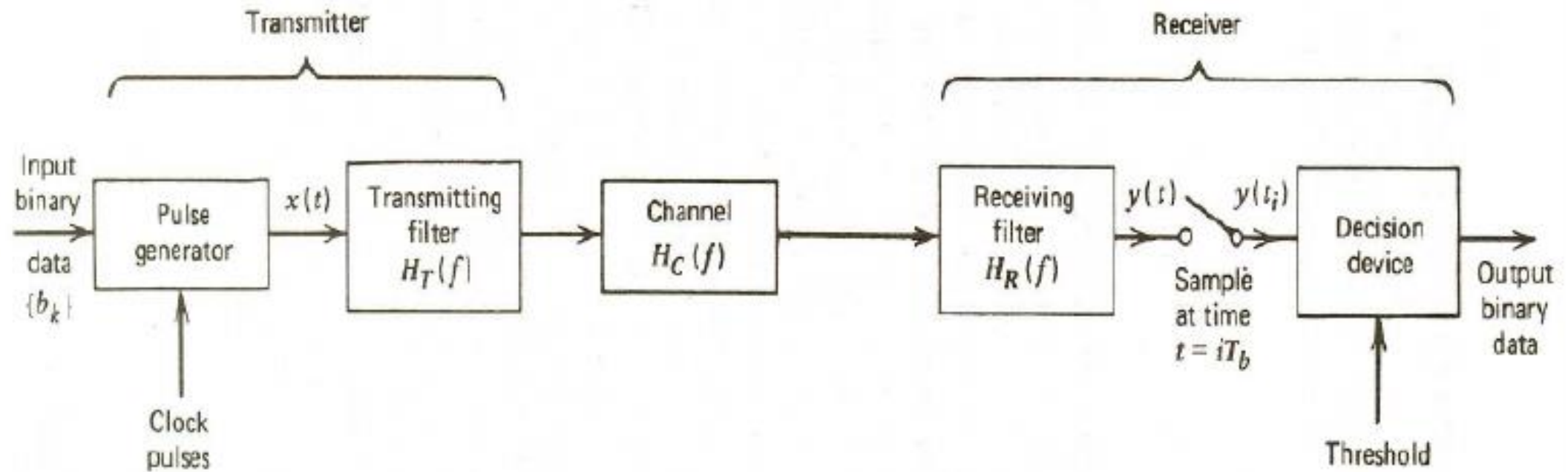
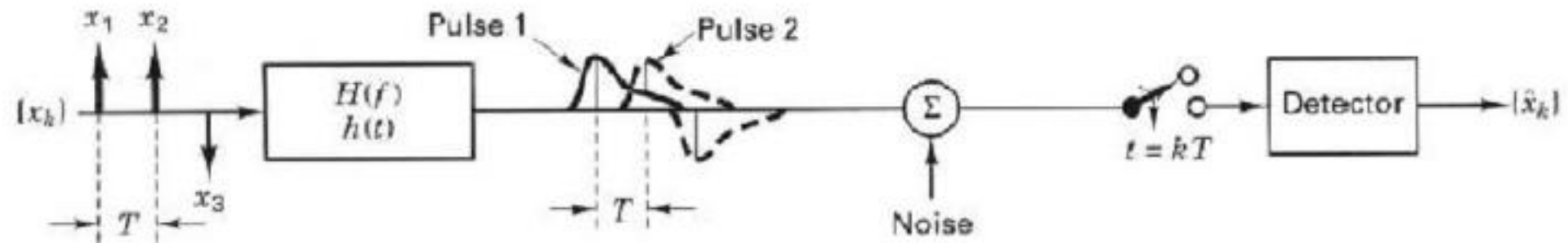
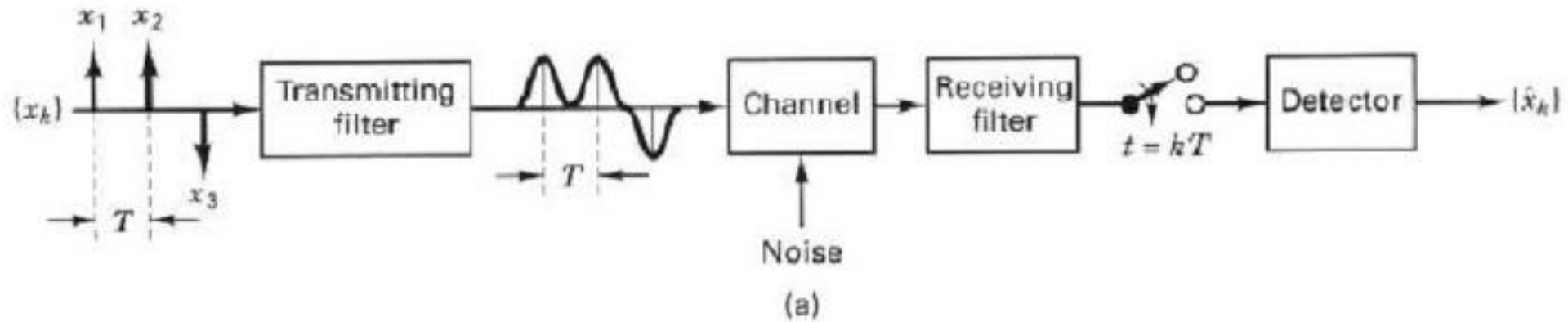


Figure 6.5 Baseband binary data transmission system.

Theory of ISI



Theory of ISI

❖ The receiving filter output may be written as

$$y(t) = \mu \sum_{k=-\infty}^{\infty} a_k p(t - kT_b)$$

- μ is scaling factor
- The pulse $p(t)$ is normalized such that

$$p(0) = 1$$

- ❖ The output $y(t)$ is produced in response to binary data waveform applied to the input of the transmitting filter. Especially, the pulse $\mu p(t)$ is response of the cascade connection of the transmitting filter, the channel, and the receiving filter, which is produced by the pulse $v(t)$ applied to the input of this cascade connection

$$\mu P(f) = V(f)H_T(f)H_C(f)H_R(f)$$

- $P(f)$ and $V(f)$ are Fourier transform of $p(t)$ and $v(t)$

❖ The receiving filter output $y(t)$ is sampled at time $t_i = iT_b$

$$\begin{aligned} y(t_i) &= \mu \sum_{k=-\infty}^{\infty} a_k p(iT_b - kT_b) \\ &= \mu a_i + \mu \sum_{\substack{k=-\infty \\ k \neq i}}^{\infty} a_k p(iT_b - kT_b) \end{aligned}$$

- ❑ The first term is produced by the i th transmitted bit.
- ❑ The second term represents the residual effect of all other transmitted bits on the decoding of the i th bit; this residual effect is called **intersymbol interference** (ISI)

Nyquist's Criterion For Distortionless (Zero ISI)

- ❖ Typically, the transfer function of the channel and the transmitted pulse shape are specified, and the problem is to determine the transfer functions of the transmitting and receiving filters so as to reconstruct the transmitted data sequence $\{b_k\}$
- ❖ The receiver does this by extracting and then decoding the corresponding sequence of weights, $\{a_k\}$, from the output $y(t)$.
- ❖ Except for a scaling factor, $y(t)$ is determined by the a_k and the received pulse $p(t)$

Nyquist's Criterion For Distortionless (Zero ISI)

- ❖ The **extraction** involves sampling the output $y(t)$ at some time $t = iT_b$
- ❖ The **decoding** requires that the weighted pulse contribution $a_k p(iT_b - kT_b)$ for $k = i$ be free from ISI due to the overlapping tails of all other weighted pulse contributions represented by $k \neq i$

Nyquist's Criterion For Distortionless (Zero ISI)

❖ This, in turn, require that we control the received pulse $p(t)$, as shown by

$$p(iT_b - kT_b) = \begin{cases} 1 & i = k \\ 0 & i \neq k \end{cases}$$

□ where, by normalization, $p(0) = 1$

Nyquist's Criterion For Distortionless (Zero ISI)

❖ The receiver output

$$y(t_i) = \mu a_i$$

- Which implies zero intersymbol interference (ISI)
- This condition assures perfect reception in the absence of noise

- ❖ Consider the sequence of samples $\{p(nT_b)\}$, where $n = 0, \pm 1, \pm 2, \dots$
- ❖ Sampling in the time domain produces periodicity in frequency domain

$$P_\delta(f) = R_b \sum_{n=-\infty}^{\infty} P(f - nR_b)$$

- Where $R_b = 1/T_b$ is the bit rate
- $P_\delta(f)$ is the Fourier transform of an infinite periodic sequence of delta functions of period T_b , and whose strengths are weighted by the respective sample values of $p(t)$

❖ That is

$$P_{\delta}(f) = \int_{-\infty}^{\infty} \left[\sum p(mT_b) \delta(t - mT_b) \right] \exp(-j2\pi ft) dt$$

- where $m = i - k$.

❖ Impose the condition of zero ISI on the sample values of $p(t)$

$$\begin{aligned} P_{\delta}(f) &= \int_{-\infty}^{\infty} p(0) \delta(t) \exp(-j2\pi ft) dt \\ &= p(0) \end{aligned}$$

❖ Since $p(0) = 1$, by normalization, the condition for zero ISI is satisfied if

$$\sum_{n=-\infty}^{\infty} P(f - nR_b) = T_b$$

→ *Nyquist criterion for distortionless baseband transmission*

Nyquist bandwidth constraint

❖ Nyquist bandwidth constraint (on equivalent system):

- The theoretical minimum required system bandwidth to detect R_s [symbols/s] without ISI is $R_s/2$ [Hz].
- Equivalently, a system with bandwidth $W=1/2T=R_s/2$ [Hz] can support a maximum transmission rate of $2W=1/T=R_s$ [symbols/s] without ISI.

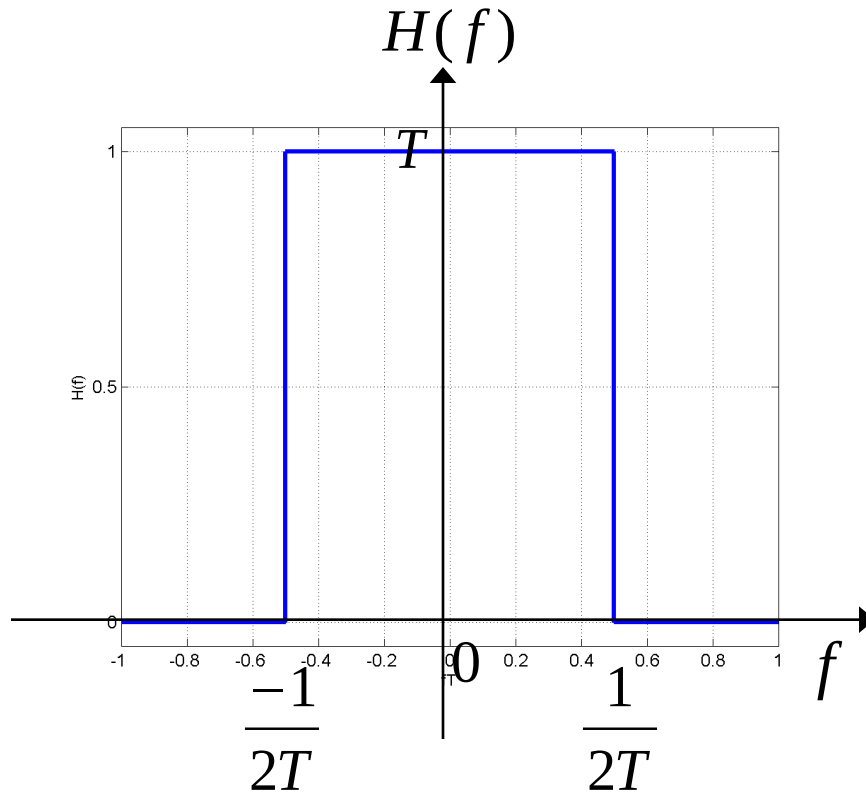
$$\frac{1}{2T} = \frac{R_s}{2} \leq W \Rightarrow \frac{R_s}{W} \geq 2 \quad [\text{symbol/s/Hz}]$$

❖ Bandwidth efficiency, R/W [bits/s/Hz] :

- An important measure in DCs representing data throughput per hertz of bandwidth.
- Showing how efficiently the bandwidth resources are used by signaling techniques.

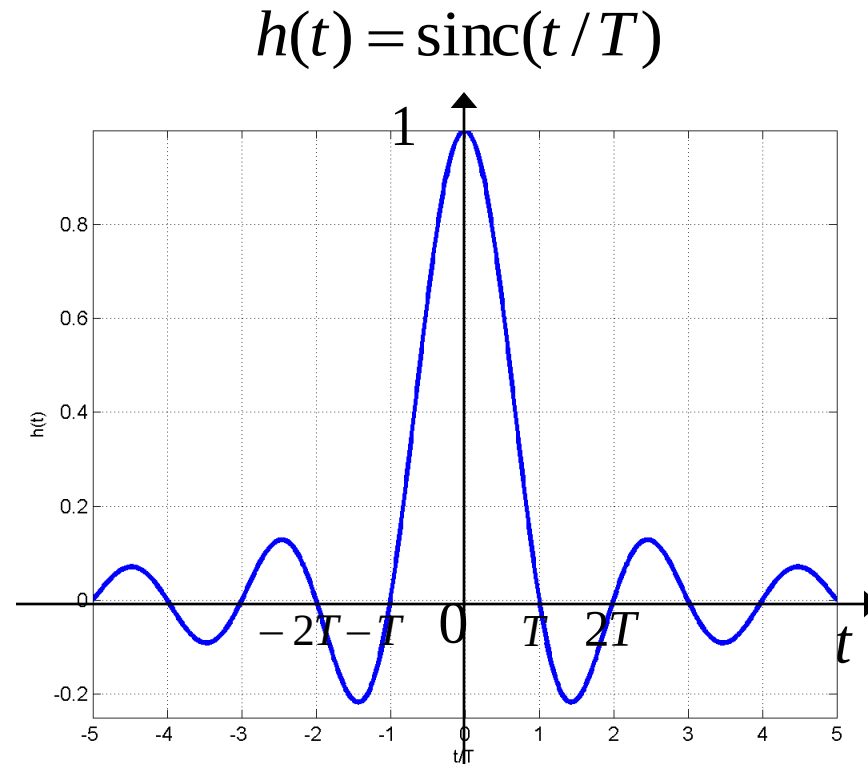
Equiv System: Ideal Nyquist pulse (filter)

Ideal Nyquist filter



$$W = \frac{1}{2T}$$

Ideal Nyquist pulse



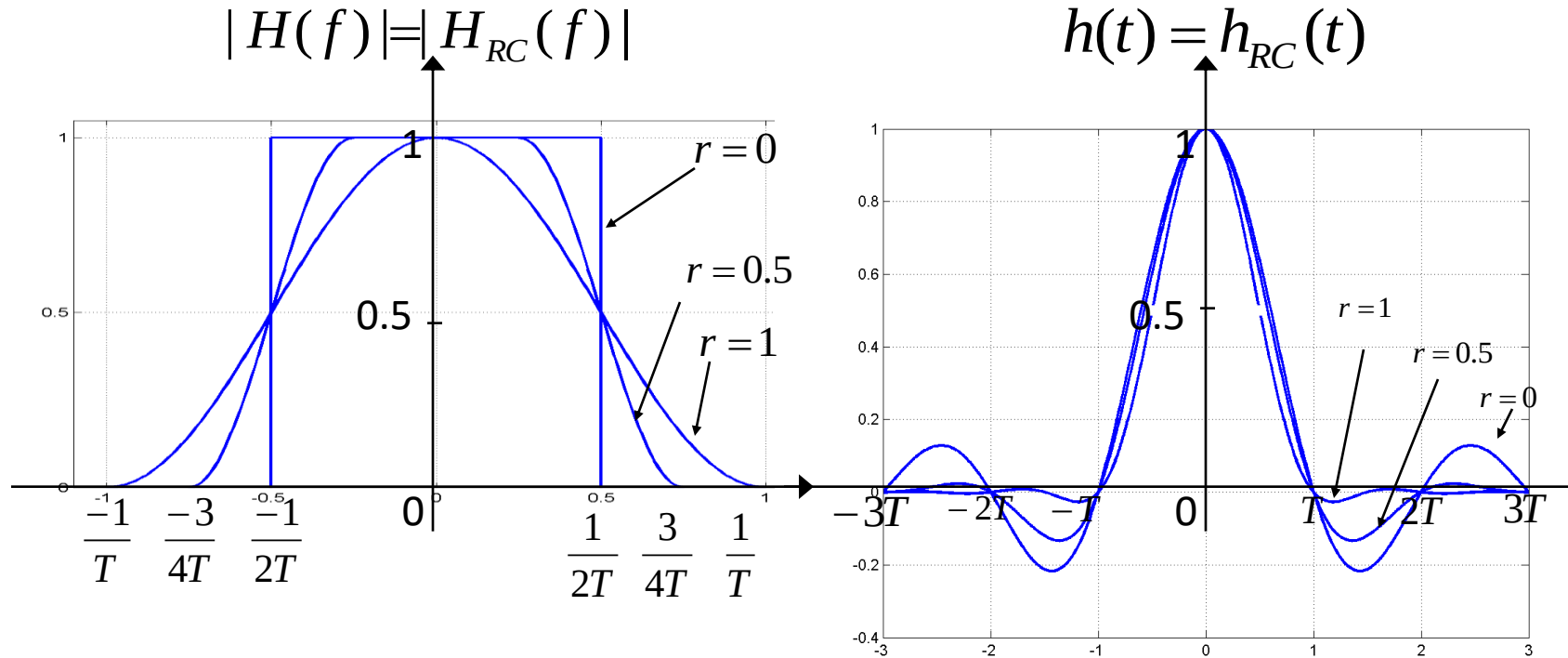
Nyquist pulses (filters)

- Nyquist pulses (filters):
 - Pulses (filters) which result in no ISI at the sampling time.
- Nyquist filter:
 - Its transfer function in frequency domain is obtained by convolving a rectangular function with any real even-symmetric frequency function
- Nyquist pulse:
 - Its shape can be represented by a $\text{sinc}(t/T)$ function multiply by another time function.
- Example of Nyquist filters: Raised-Cosine filter

Pulse shaping to reduce ISI

- Goals and trade-off in pulse-shaping
 - Reduce ISI
 - Efficient bandwidth utilization
 - Robustness to timing error (small side lobes)

Raised Cosine Filter: Nyquist Pulse Approximation



Baseband $W_{\text{SSB}} = (1+r) \frac{R_s}{2}$

Passband $W_{\text{DSB}} = (1+r)R_s$

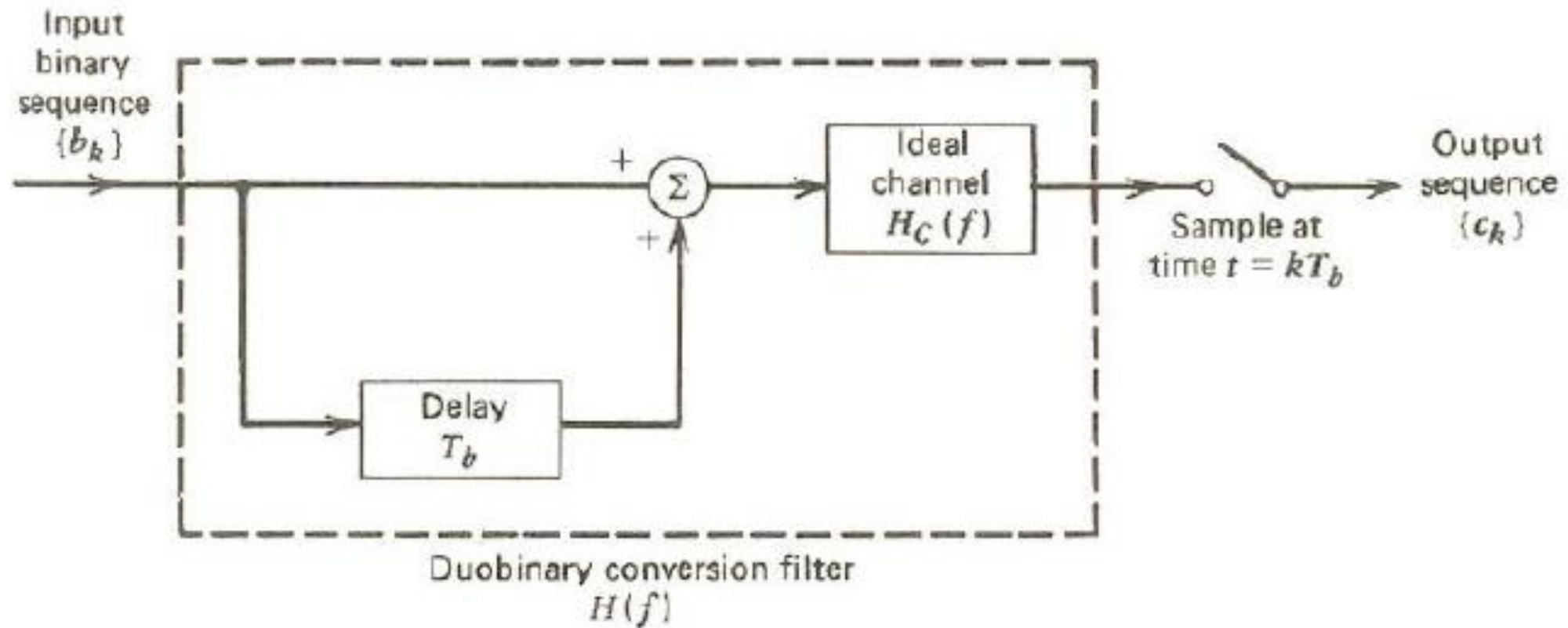
Correlative Coding

- ❖ It is possible to achieve a bit rate of $2B_0$ per second in a channel of bandwidth B_0 Hertz by adding intersymbol interference to the transmitted signal in a controlled manner
- ❖ Such schemes are called correlative coding or partial-response signaling schemes
- ❖ The design of these schemes is based on the premise that since intersymbol interference introduced into the transmitted signal is known, its effect can be compensated at the receiver.
- ❖ Correlative coding may be regarded as a practical means of achieving the theoretical maximum signaling rate of $2B_0$ per second in a bandwidth of B_0 hertz

❖ Duobinary signaling

- Consider a binary input sequence $\{b_k\}$ consisting of uncorrelated binary digits each having duration T_b seconds, with symbol 1 represented by a pulse of amplitude +1 volt, and symbol 0 by a pulse of amplitude -1 volt
- This sequence is applied to **duobinary encoder**, it is converted into a three-level output, namely -2, 0, and +2 volts

❖ Duobinary signaling



❖ Duobinary signaling

- The digit c_k at the duobinary coder output is the sum of the resent binary digit b_k and its previous value b_{k-1}

$$c_k = b_k + b_{k-1}$$

- One of the effects of the transformation is to change the input sequence $\{b_k\}$ of uncorrelated binary digits into a sequence $\{c_k\}$ of correlated digits
- This correlation between the adjacent transmitted levels may be viewed as introducing ISI into the transmitted signal

❖ Duobinary signaling

- The overall transfer function of this filter connected in cascade with the ideal channel $H_c(f)$ is

$$\begin{aligned} H(f) &= H_c(f) \left[1 + \exp(-j2\pi f T_b) \right] \\ &= H_c(f) \left[1 + \exp(j\pi f T_b) + \exp(-j\pi f T_b) \right] \exp(-j\pi f T_b) \\ &= 2H_c(f) \cos(\pi f T_b) \exp(-j\pi f T_b) \end{aligned}$$

❖ Duobinary signaling

- For the ideal channel of bandwidth $B_0 = R_b/2$, we have

$$H_c(f) = \begin{cases} 1 & |f| \leq R_b/2 \\ 0 & \text{otherwise} \end{cases}$$

- The overall frequency response has the form of a half-cycle cosine function

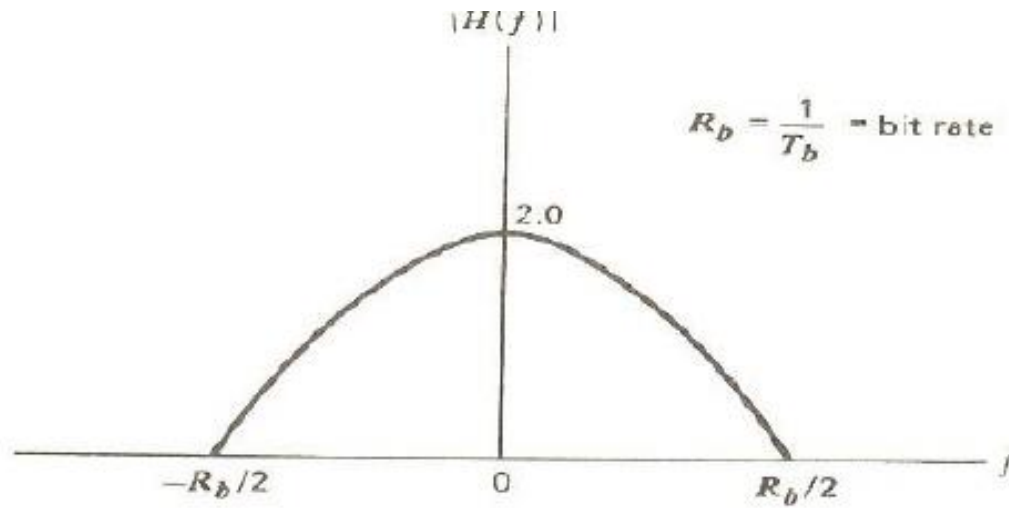
$$H(f) = \begin{cases} 2 \cos(\pi f T_b) \exp(-j \pi f T_b) & |f| \leq R_b/2 \\ 0 & \text{otherwise} \end{cases}$$

❖ Duobinary signaling

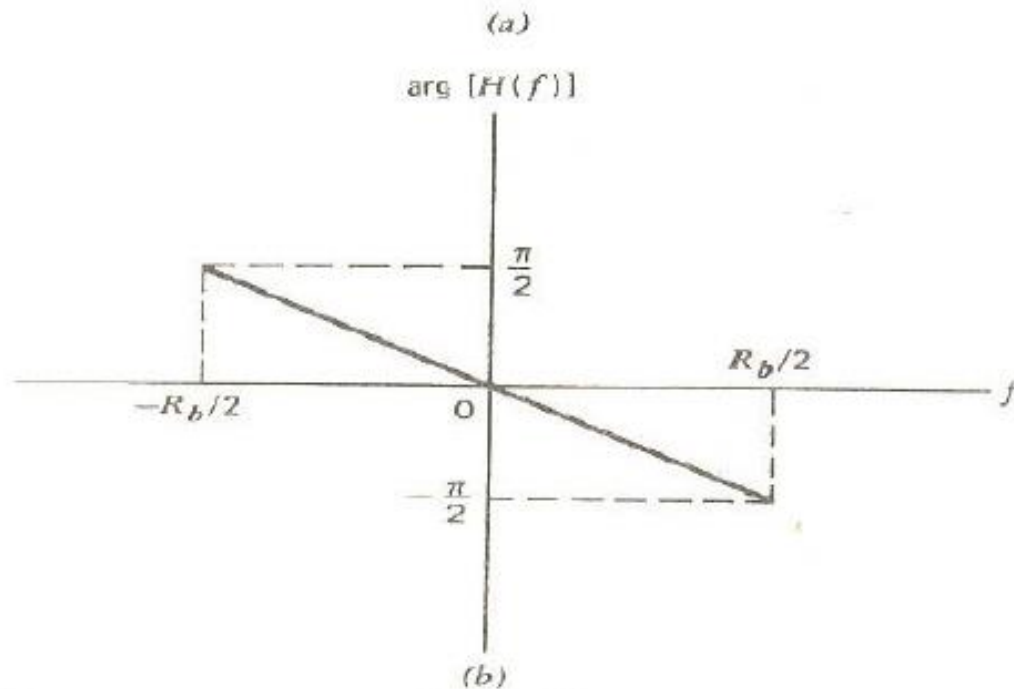
- The corresponding value of the impulse response consists of two sinc pulse, time-displaced by T_b seconds

$$\begin{aligned} h(t) &= \frac{\sin(\pi t/T_b)}{\pi t/T_b} + \frac{\sin[\pi(t - T_b)/T_b]}{\pi(t - T_b)/T_b} \\ &= \frac{\sin(\pi t/T_b)}{\pi t/T_b} - \frac{\sin[\pi t/T_b]}{\pi(t - T_b)/T_b} \\ &= \frac{T_b^2 \sin(\pi t/T_b)}{\pi t(T_b - t)} \end{aligned}$$

Frequency response of duobinary conversion filter



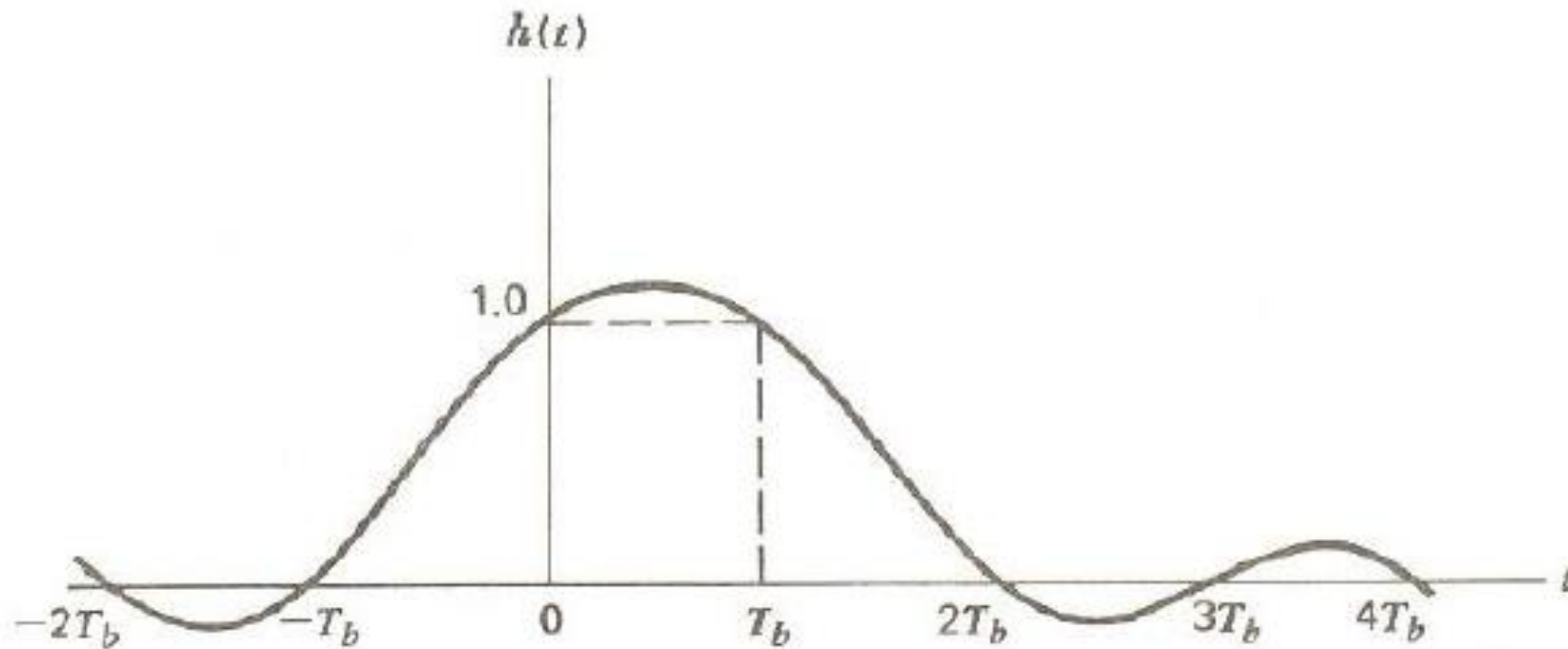
Amplitude Response



Phase Response

Impulse response of duobinary conversion filter

❖ Duobinary signaling



Baseband M-ary PAM

- ❖ Consider more of pulse amplitude modulation.
- ❖ It arises at some point in almost all systems
- ❖ Its performance and design is key to many systems.

Baseband M-ary PAM

- Consider a transmitted signal of the form

$$s(t) = \sum_k a_k p(t - kT), \text{ where } p(t) \text{ is usually a Nyquist or root}$$

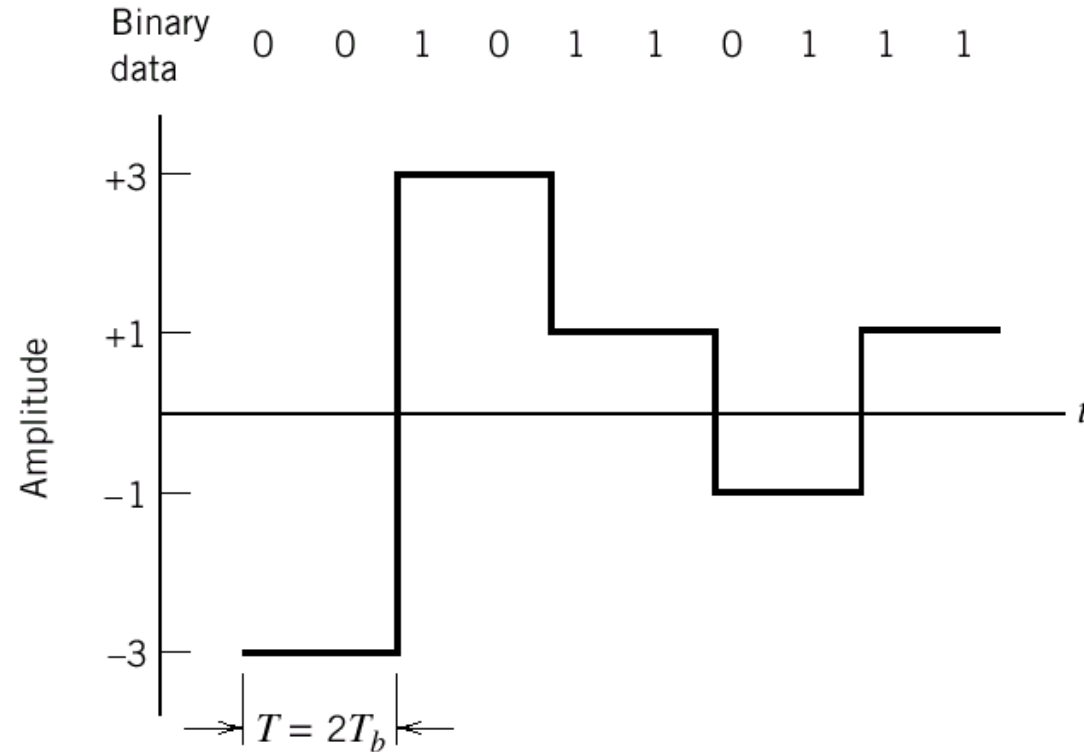
Nyquist pulse shape.

- Consider here that symbols a_k take on $M > 2$ values

Eg. $a_k \in \{-3, -1, +1, +3\}$

- T^{-1} is the signaling rate in symbols/sec (baud)
- Assume Gray coding of bits to symbols
- $T = T_b \log_2 M$, where T_b = bit interval.
- Such M -ary symbols use less bandwidth, but more power than a binary signal for a given performance level.
- Illustrated in the following for $M = 4$.

Baseband M-ary PAM



(a)

Dibit	Amplitude
00	-3
01	-1
11	+1
10	+3

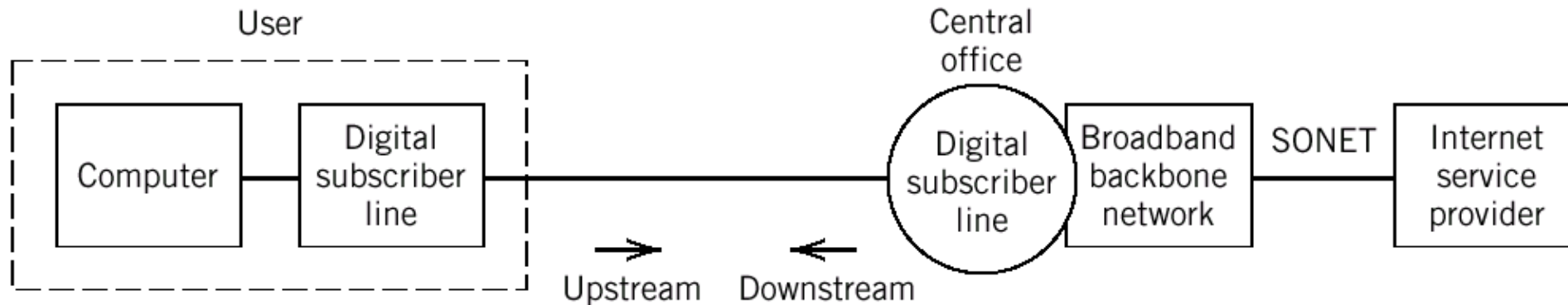
(b)

- Illustrates 4-level PAM using NRZ pulses.
- In practice these are then passed through a shaping filter to obtain Nyquist pulses.

Baseband M-ary PAM

- **Digital Subscriber Lines (DSL):** Major Application of PAM
- Use normal twisted pair that historically has been used for POTS.
- Actually bandpass signals, but can consider as PAM since modulation is simple frequency translation.
- Used in local loop typically at distances of < 1.5 km to exchange.
- Because of band limited nature of telephone channel, ISI is a major problem and must be compensated.

Operational Environment:

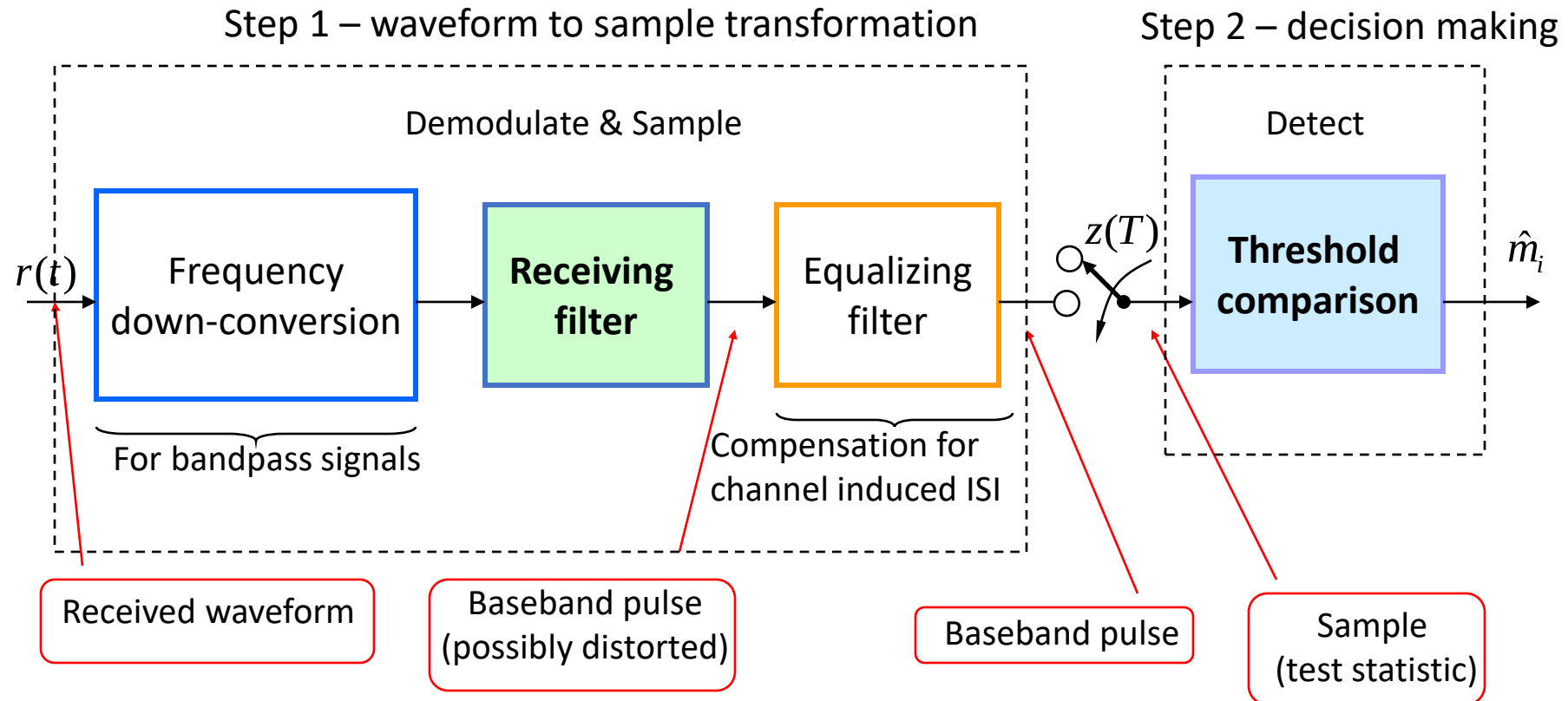


Are several different DSL modes.

optimum linear receiver

- Demodulation and sampling:
 - Waveform recovery and preparing the received signal for detection:
 - Improving the signal power to the noise power (SNR) using matched filter (project to signal space)
 - Reducing ISI using equalizer (remove channel distortion)
 - Sampling the recovered waveform
- Detection:
 - Estimate the transmitted symbol based on the received sample

Receiver structure



Equalization

Definition of the Equalizer: the equalizer is a digital filter that provides an approximate inverse of channel frequency response.

Goal of equalization: is to mitigate the effects of ISI to decrease the probability of error that occurs without suppression of ISI, but this reduction of ISI effects has to be balanced with the prevention of noise power enhancement.

Types of Equalization techniques

Linear Equalization techniques which are simple to implement, but greatly enhance noise power because they work by inverting channel frequency response.

Non-Linear Equalization techniques which are more complex to implement, but have much less noise enhancement than linear equalizers.

Thank You